



PREALG: Basic Geometry

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Learning Objectives

- Compute the perimeter of rectangles and polygons
- Compute the area of rectangles and triangles
- Identify and classify angles (acute, right, obtuse)
- Apply geometric formulas to real-world problems

Simplify each expression completely. Show all steps and circle your final answer.

Area of a rectangle

1. Find the area of a rectangle with length 13 ft and width 3 ft.

$$l = 13, w = 3$$

Answer: _____

2. A garden measures 5 meters long and 6 meters wide. How many square meters of soil are needed to cover it?

$$l = 5, w = 6$$

Answer: _____

3. A rectangular parking lot is 41 yards long and 26 yards wide. What is the area of the lot?

$$l = 41, w = 26$$

Answer: _____

4. Find the area of a rectangle with length 7 ft and width 2 ft.

$$l = 7, w = 2$$

Answer: _____

5. A garden measures 10 meters long and 9 meters wide. How many square meters of soil are needed to cover it?

$$l = 10, w = 9$$

Answer: _____



6. A rectangular parking lot is 23 yards long and 24 yards wide. What is the area of the lot?

$$l = 23, w = 24$$

Answer: _____

7. Find the area of a rectangle with length 12 ft and width 4 ft.

$$l = 12, w = 4$$

Answer: _____

8. A garden measures 18 meters long and 9 meters wide. How many square meters of soil are needed to cover it?

$$l = 18, w = 9$$

Answer: _____

9. A rectangular parking lot is 45 yards long and 13 yards wide. What is the area of the lot?

$$l = 45, w = 13$$

Answer: _____

Area of a triangle

10. Find the area of a triangle with base 10 cm and height 8 cm.

$$b = 10, h = 8$$

Answer: _____

11. Find the area of a triangle with base 5 m and height 9 m.

$$b = 5, h = 9$$

Answer: _____

12. A triangular sail has a base of 11 ft and a height of 5 ft. What is the area of the sail?

$$b = 11, h = 5$$

Answer: _____

13. Find the area of a triangle with base 4 cm and height 5 cm.

$$b = 4, h = 5$$

Answer: _____



14. Find the area of a triangle with base 12 m and height 3 m.

$$b = 12, h = 3$$

Answer: _____

15. A triangular sail has a base of 6 ft and a height of 9 ft. What is the area of the sail?

$$b = 6, h = 9$$

Answer: _____

16. Find the area of a triangle with base 14 cm and height 6 cm.

$$b = 14, h = 6$$

Answer: _____

17. Find the area of a triangle with base 5 m and height 7 m.

$$b = 5, h = 7$$

Answer: _____

18. A triangular sail has a base of 7 ft and a height of 5 ft. What is the area of the sail?

$$b = 7, h = 5$$

Answer: _____

Perimeter of a rectangle

19. Find the perimeter of a rectangle with length 13 cm and width 4 cm.

$$l = 13, w = 4$$

Answer: _____

20. A rectangular room is 9 feet long and 11 feet wide. How much baseboard trim (in feet) is needed to go around the entire room?

$$l = 9, w = 11$$

Answer: _____

21. Find the perimeter of a rectangle with length 9 cm and width 5 cm.

$$l = 9, w = 5$$

Answer: _____



22. A rectangular room is 11 feet long and 7 feet wide. How much baseboard trim (in feet) is needed to go around the entire room?

$$l = 11, w = 7$$

Answer: _____

23. Find the perimeter of a rectangle with length 17 cm and width 10 cm.

$$l = 17, w = 10$$

Answer: _____

24. A rectangular room is 18 feet long and 6 feet wide. How much baseboard trim (in feet) is needed to go around the entire room?

$$l = 18, w = 6$$

Answer: _____

Perimeter of a triangle

25. Find the perimeter of a triangle with sides 10 cm, 7 cm, and 8 cm.

$$a = 10, b = 7, c = 8$$

Answer: _____

26. A triangular garden has sides measuring 8 ft, 18 ft, and 13 ft. How many feet of fencing are needed to enclose it?

$$a = 8, b = 18, c = 13$$

Answer: _____

27. Find the perimeter of a triangle with sides 7 cm, 7 cm, and 12 cm.

$$a = 7, b = 7, c = 12$$

Answer: _____

28. A triangular garden has sides measuring 12 ft, 7 ft, and 13 ft. How many feet of fencing are needed to enclose it?

$$a = 12, b = 7, c = 13$$

Answer: _____

29. Find the perimeter of a triangle with sides 4 cm, 7 cm, and 14 cm.

$$a = 4, b = 7, c = 14$$

Answer: _____



30. A triangular garden has sides measuring 10 ft, 13 ft, and 14 ft. How many feet of fencing are needed to enclose it?

$$a = 10, b = 13, c = 14$$

Answer: _____





Topics: Area of a rectangle, Perimeter of a triangle, Area of a triangle, Perimeter of a rectangle. All answers verified by independent computation.

Solutions



Area of a rectangle

1. Find the area of a rectangle with length 13 ft and width 3 ft.

$$l = 13, w = 3$$

$$\rightarrow A = l \times w = 13 \times 3 = 39 \text{ sq ft.}$$

Answer: $A = 13 \times 3 = 39$

2. A garden measures 5 meters long and 6 meters wide. How many square meters of soil are needed to cover it?

$$l = 5, w = 6$$

$$\rightarrow \text{Area} = 5 \times 6 = 30 \text{ square meters.}$$

Answer: $A = 5 \times 6 = 30$

3. A rectangular parking lot is 41 yards long and 26 yards wide. What is the area of the lot?

$$l = 41, w = 26$$

$$\rightarrow \text{Area} = \text{length} \times \text{width.}$$

$$\rightarrow A = 41 \times 26 = 1066 \text{ yd}^2.$$

Answer: $A = 41 \times 26 = 1066$

4. Find the area of a rectangle with length 7 ft and width 2 ft.

$$l = 7, w = 2$$

$$\rightarrow A = l \times w = 7 \times 2 = 14 \text{ sq ft.}$$

Answer: $A = 7 \times 2 = 14$

5. A garden measures 10 meters long and 9 meters wide. How many square meters of soil are needed to cover it?

$$l = 10, w = 9$$

$$\rightarrow \text{Area} = 10 \times 9 = 90 \text{ square meters.}$$

Answer: $A = 10 \times 9 = 90$

6. A rectangular parking lot is 23 yards long and 24 yards wide. What is the area of the lot?

$$l = 23, w = 24$$

$$\rightarrow \text{Area} = \text{length} \times \text{width.}$$

$$\rightarrow A = 23 \times 24 = 552 \text{ yd}^2.$$

Answer: $A = 23 \times 24 = 552$

7. Find the area of a rectangle with length 12 ft and width 4 ft.

$$l = 12, w = 4$$

$$\rightarrow A = l \times w = 12 \times 4 = 48 \text{ sq ft.}$$

Answer: $A = 12 \times 4 = 48$



8. A garden measures 18 meters long and 9 meters wide. How many square meters of soil are needed to cover it?

$$l = 18, w = 9$$

$$\rightarrow \text{Area} = 18 \times 9 = 162 \text{ square meters.}$$

Answer: $A = 18 \times 9 = 162$

9. A rectangular parking lot is 45 yards long and 13 yards wide. What is the area of the lot?

$$l = 45, w = 13$$

$$\rightarrow \text{Area} = \text{length} \times \text{width.}$$

$$\rightarrow A = 45 \times 13 = 585 \text{ yd}^2.$$

Answer: $A = 45 \times 13 = 585$



Area of a triangle

10. Find the area of a triangle with base 10 cm and height 8 cm.

$$b = 10, h = 8$$

$$\rightarrow A = (1/2) \times \text{base} \times \text{height} = (1/2) \times 10 \times 8 = 80/2 = 40 \text{ sq cm.}$$

Answer: $A = \frac{1}{2} \times 10 \times 8 = \frac{80}{2} = 40$

11. Find the area of a triangle with base 5 m and height 9 m.

$$b = 5, h = 9$$

$$\rightarrow \text{Formula: } A = (1/2) \times \text{base} \times \text{height.}$$

$$\rightarrow A = (1/2) \times 5 \times 9 = 45/2 = 22 \text{ m}^2.$$

Answer: $A = \frac{1}{2} \times 5 \times 9 = \frac{45}{2} = 22$

12. A triangular sail has a base of 11 ft and a height of 5 ft. What is the area of the sail?

$$b = 11, h = 5$$

$$\rightarrow A = (1/2) \times \text{base} \times \text{height} = (1/2) \times 11 \times 5.$$

$$\rightarrow A = 55/2 = 27 \text{ ft}^2.$$

Answer: $A = \frac{1}{2} \times 11 \times 5 = \frac{55}{2} = 27$

13. Find the area of a triangle with base 4 cm and height 5 cm.

$$b = 4, h = 5$$

$$\rightarrow A = (1/2) \times \text{base} \times \text{height} = (1/2) \times 4 \times 5 = 20/2 = 10 \text{ sq cm.}$$

Answer: $A = \frac{1}{2} \times 4 \times 5 = \frac{20}{2} = 10$

14. Find the area of a triangle with base 12 m and height 3 m.

$$b = 12, h = 3$$

$$\rightarrow \text{Formula: } A = (1/2) \times \text{base} \times \text{height.}$$

$$\rightarrow A = (1/2) \times 12 \times 3 = 36/2 = 18 \text{ m}^2.$$

Answer: $A = \frac{1}{2} \times 12 \times 3 = \frac{36}{2} = 18$

15. A triangular sail has a base of 6 ft and a height of 9 ft. What is the area of the sail?

$$b = 6, h = 9$$

$$\rightarrow A = (1/2) \times \text{base} \times \text{height} = (1/2) \times 6 \times 9.$$

$$\rightarrow A = 54/2 = 27 \text{ ft}^2.$$

Answer: $A = \frac{1}{2} \times 6 \times 9 = \frac{54}{2} = 27$



16. Find the area of a triangle with base 14 cm and height 6 cm.

$$b = 14, h = 6$$

$$\rightarrow A = (1/2) \times \text{base} \times \text{height} = (1/2) \times 14 \times 6 = 84/2 = 42 \text{ sq cm.}$$

Answer: $A = \frac{1}{2} \times 14 \times 6 = \frac{84}{2} = 42$

17. Find the area of a triangle with base 5 m and height 7 m.

$$b = 5, h = 7$$

$$\rightarrow \text{Formula: } A = (1/2) \times \text{base} \times \text{height.}$$

$$\rightarrow A = (1/2) \times 5 \times 7 = 35/2 = 17 \text{ m}^2.$$

Answer: $A = \frac{1}{2} \times 5 \times 7 = \frac{35}{2} = 17$

18. A triangular sail has a base of 7 ft and a height of 5 ft. What is the area of the sail?

$$b = 7, h = 5$$

$$\rightarrow A = (1/2) \times \text{base} \times \text{height} = (1/2) \times 7 \times 5.$$

$$\rightarrow A = 35/2 = 17 \text{ ft}^2.$$

Answer: $A = \frac{1}{2} \times 7 \times 5 = \frac{35}{2} = 17$



Perimeter of a rectangle

19. Find the perimeter of a rectangle with length 13 cm and width 4 cm.

$$l = 13, w = 4$$

$$\rightarrow P = 2(l + w) = 2(13 + 4) = 2(17) = 34 \text{ cm.}$$

Answer: $P = 2(13 + 4) = 2(17) = 34$

20. A rectangular room is 9 feet long and 11 feet wide. How much baseboard trim (in feet) is needed to go around the entire room?

$$l = 9, w = 11$$

$$\rightarrow \text{Perimeter} = 2(\text{length} + \text{width}).$$

$$\rightarrow P = 2(9 + 11) = 2(20) = 40 \text{ ft.}$$

Answer: $P = 2(9 + 11) = 2(20) = 40$

21. Find the perimeter of a rectangle with length 9 cm and width 5 cm.

$$l = 9, w = 5$$

$$\rightarrow P = 2(l + w) = 2(9 + 5) = 2(14) = 28 \text{ cm.}$$

Answer: $P = 2(9 + 5) = 2(14) = 28$

22. A rectangular room is 11 feet long and 7 feet wide. How much baseboard trim (in feet) is needed to go around the entire room?

$$l = 11, w = 7$$

$$\rightarrow \text{Perimeter} = 2(\text{length} + \text{width}).$$

$$\rightarrow P = 2(11 + 7) = 2(18) = 36 \text{ ft.}$$

Answer: $P = 2(11 + 7) = 2(18) = 36$

23. Find the perimeter of a rectangle with length 17 cm and width 10 cm.

$$l = 17, w = 10$$

$$\rightarrow P = 2(l + w) = 2(17 + 10) = 2(27) = 54 \text{ cm.}$$

Answer: $P = 2(17 + 10) = 2(27) = 54$

24. A rectangular room is 18 feet long and 6 feet wide. How much baseboard trim (in feet) is needed to go around the entire room?

$$l = 18, w = 6$$

$$\rightarrow \text{Perimeter} = 2(\text{length} + \text{width}).$$

$$\rightarrow P = 2(18 + 6) = 2(24) = 48 \text{ ft.}$$

Answer: $P = 2(18 + 6) = 2(24) = 48$



Perimeter of a triangle

25. Find the perimeter of a triangle with sides 10 cm, 7 cm, and 8 cm.

$$a = 10, b = 7, c = 8$$

→ Perimeter = sum of all three sides.

$$\rightarrow P = 10 + 7 + 8 = 25 \text{ cm.}$$

Answer: $P = 10 + 7 + 8 = 25$

26. A triangular garden has sides measuring 8 ft, 18 ft, and 13 ft. How many feet of fencing are needed to enclose it?

$$a = 8, b = 18, c = 13$$

→ Add all three sides: $8 + 18 + 13 = 39$.

→ You need 39 ft of fencing.

Answer: $P = 8 + 18 + 13 = 39$

27. Find the perimeter of a triangle with sides 7 cm, 7 cm, and 12 cm.

$$a = 7, b = 7, c = 12$$

→ Perimeter = sum of all three sides.

$$\rightarrow P = 7 + 7 + 12 = 26 \text{ cm.}$$

Answer: $P = 7 + 7 + 12 = 26$

28. A triangular garden has sides measuring 12 ft, 7 ft, and 13 ft. How many feet of fencing are needed to enclose it?

$$a = 12, b = 7, c = 13$$

→ Add all three sides: $12 + 7 + 13 = 32$.

→ You need 32 ft of fencing.

Answer: $P = 12 + 7 + 13 = 32$

29. Find the perimeter of a triangle with sides 4 cm, 7 cm, and 14 cm.

$$a = 4, b = 7, c = 14$$

→ Perimeter = sum of all three sides.

$$\rightarrow P = 4 + 7 + 14 = 25 \text{ cm.}$$

Answer: $P = 4 + 7 + 14 = 25$

30. A triangular garden has sides measuring 10 ft, 13 ft, and 14 ft. How many feet of fencing are needed to enclose it?

$$a = 10, b = 13, c = 14$$

→ Add all three sides: $10 + 13 + 14 = 37$.

→ You need 37 ft of fencing.

Answer: $P = 10 + 13 + 14 = 37$

